INCH-POUND

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MILITARY SPECIFICATION

TRANSMITTER, PRESSURE TRU-81/A, OIL, 0-500 PSI

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote-indicating, 320° movement, synchro style, differential, oil-pressure transmitter, designated TRU-81/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, Standards, and Handbooks. The following specifications standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-3037 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document of by letter.

AMSC F3828 FSC 6620

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FEDERAL

QQ-S-763	Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting
PPP-B-636	Box, Shipping Fiberboard Cushioning

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-B-131	Barrier Material, Water-proof, Greaseproof, Flexible, Heat-Sealable
MIL-E-5400	Electronic Equipment, Aerospace, General Specification For
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-I-6856	Indicator, Pressure, Synchro, 1-1/2 Inch
MIL-I-7057	Indicator, Synchro, Aircraft, General Specification For
MIL-S-7742	Screw Threads Standard, Optimum Selected Series, General Specification For
MIL-A-8625	Anodic Coatings, Aluminum and Aluminum Alloys
MIL-S-20708	Synchros, General Specification For
MIL-B-27497	Bearing, Jewel, Sapphire Or Ruby, Synthetic

STANDARDS

FEDERAL

FED-STD-101	Test Procedures For Packaging Materials

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U S Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards & Specifications, Order Of Preference For The Selection Of

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MIL-STD-2073-1A	DOD Material Procedures For Development & Application Of Packaging Requirements
MS21209	Insert, Screw Thread, Coarse And Fine, Screw Locking, Helical Coil, Cres
MS33649	Bosses, Fluid Connection Internal Straight Thread
MS33678	Connector, Receptacle, Electric, Integral Mounting

MILITARY

MIL-HDBK-275	Guide For Selection Of Lubricants, Fluids, And Compounds For Use In Flight Vehicles And Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094).

2.2 Order of precedence. In the event of a conflict between the test of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Transmitters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.4).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials. Materials used in the manufacture of transmitters shall be suitable for the purpose and shall conform to applicable Government specifications. Materials conforming to contractor's specification may be used provided such specifications are approved by the Government and contain provisions for adequate tests. The use of the contractor's specification will not constitute waiver of Government inspection.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically
sealed,they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are used in a hermetically sealed enclosure, fungicidal treatment will not necessary.

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3.3.2 Recovered Materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.3.4 Dissimilar metals. Unless suitable protect against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.3.6 Corrosive fumes. Materials used in the construction of the transmitter shall not produce corrosive or deleterious fumes under test conditions specified herein.

3.3.7 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction

3.4.1 Design. The transmitter shall be designed to operate with indicators conforming to MIL-I-6856 and MIL-I-7057.

3.4.1.1 Outline dimension. The outline dimensions of the transmitter shall conform to Figure 1.

3.4.1.2 Adjustments. The transmitter shall be provided with a means for adjusting or correcting indications which shall be simple to manipulate and such that adjustments can be accomplished with tools ordinarily available. The means of adjustment shall be accessible from the exterior of the case and shall provide a minimum of ±7 angular degrees of synchro rotor adjustment from electrical zero.

3.4.1.3 Mechanisms. Transmitter mechanisms shall be comprised of units distinct from their cases. The units shall be sufficiently rugged so that small distortions of the cases will not affect the accuracy of the indications.

3.4.1.4 Adjustment locks. All calibration adjustments shall be provided with securing devices to lock the adjusting mechanism in place to prevent the adjustment from drifting from its original setting due to environmental conditions specified herein. Provisions for locking shall be made on the calibration adjustment so that it can not be readily moved.

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3.4.1.4 Reliability. The transmitter shall have specified mean-time-between-failures (MBTF) of 2,000 hours at a 90 percent confidence level when tested as specified under 4.6.23.

3.4.1.6 Lubricant. The contractor shall select a lubricant, or lubricants, in accordance with MIL-HDBK-275 which shall be suitable for use on gears, joints,
bearings, and pinions of the transmitter. Prior to selecting the lubricant or lubricants, the contractor shall make an investigation relative to the lubricant
most suitable for the application with due consideration being given to the environmental conditions under which the transmitter must operate.

3.4.1.7 Synchro transmitters. The synchro transmitter component of the pressure transmitter shall meet all performance requirements of MIL-S-20708 except as specified herein.

3.4.1.7.1 Synchro bearings. The precision ball bearings used in the synchro shall be corrosion-resistant alloy steel (stainless steel) conforming to QQ-S-763, class 400C, and shall be made to ABEC tolerance 5P.

3.4.1.7.1 Synchro zero. A means shall be provided for readily setting the synchro at electrical zero. The Synchro transmitter shall be so designed that the synchro electrical zero position can be set and locked in position so securely that it will remain in that position for the service life of the transmitter or until reset.

3.4.1.8 Overpressure stop. An overpressure stop shall be provided to restrain the prime mover. The stop shall permit the transmitter rotor to rotate 12° to
± 3 beyond the 500-psi position. The stop shall not be damaged by application of 500-psi.

3.4.1.9 Gears, pivots, joints, and bearings. Before finalizing the design of the transmitter mechanism, the contractor shall investigate and determine the best known types and the proper processing of interfacing materials for use in the gears, pivots, joints, and bearings in the train of moving parts of the transmitter. These interface materials shall be selected for their ability to resist wear, corrosion, and friction when interfacing with one another. The interfacing materials shall not be dissimilar as described in MIL-STD-889. After selection of the material types and processing procedures, the contractor shall submit a description of each interfacing material and process proposed for the train of moving parts (gears, pivots, joints, bearings) and present his reason for selection of each type to the procuring activity for approval. (See 6.3)

3.4.1.9.1 Large-area interface surface. The gears, pivots, joints, and bearings shall have substantially large interface surfaces for the purpose of minimizing the effects of wear upon the calibration and accuracy of the transmitter. Single-point type surfaces, such as attained by pivots made of two perpendicular cylinders, will not be acceptable; interface surfaces made up of two parallel cylinders or curved surfaces will be acceptable if sufficient interface area is provided to minimize the effects of wear.

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3.4.1.10 Attitude. The transmitter shall be so designed that it will operate as specified herein in any position.

3.4.1.11 Connections. The transmitter shall be a differential type transmitter having a pressure port connection and separate vent port connection as shown in Figure 1.

3.4.1.12 Containment. There shall be no flow path between the two port connections, or either port connection and the outside of the transmitter. Except for the pressure and vent ports, no single transmitter failure shall result in such a path.

3.4.2 Construction. The transmitter shall be so constructed that no parts will work lose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. Pivots, bearings, and gears shall not bind nor shake and shall be as nearly frictionless as possible.

3.4.2.1 Mounting provisions. The mounting flange of the transmitter shall be constructed in accordance with Figure 1. The four mounting holes shall be provided with screw locking helical coils in accordance with MS21209 inserted from the electrical connector side of the transmitter. The helical coils shall accept and lock #6-32 threaded screws.

3.5 Performance. The transmitter shall be designed and calibrated to operate continuously and produce its greatest accuracy at a normal operating temperature of 165°C. The transmitter shall withstand the following condition:

a. Temperature: Operating temperatures ranging from -54° to +165°C and exposure to temperatures ranging from -62° to +232°C

b. Vibration: of 20g

c. Altitude-temperature: Absolute pressure equivalent to an 85,00-foot altitude at -54°C

d. Salt fog: Exposure to salt sea atmosphere for 48 hours

e. Overpressure: An overpressure of 500 psi for a 10-minute period.

3.5.1 Dielectric strength. The transmitter shall withstand 200V at a commercial frequency applied between any pin of the electrical connector not connected to the case and the case for a minimum period of 5 seconds.

3.5.2 Voltage and frequency variation. The transmitter shall withstand voltage and frequency variations from 22 to 30V and 380 to 420 Hz or any combinations thereof.

3.5.3 Electromagnetic interference. The transmitter shall meet the electromagnetic interference requirements of MIL-STD-461 for class A1 equipment.

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3.5.4 Damping. When the transmitter is subjected to an applied oil pressure sufficient to produce and indication of 500 PSI and the pressure is released from 500 to 100 PSI within 0.1 seconds, the corresponding time required for transmitter indication to change from 500 PSI to 100 PSI shall be between 3 and 7 seconds. There shall be no damage to the instrument as a result of the pressure change.

3.5.5 Indicator operation. The transmitter shall be capable of operating either one or two indicators with size 10 synchros without a change in output equivalent to more than 1° of arc.

3.5.6 Pressure bosses and electrical connector. The pressure bosses and electrical connector shall withstand an applied force of 25 foot-pounds bending moment with no evidence of damage to welds or soldering, distortion of base plates, and without the calibration shifting out of tolerance.

3.5.7 Scale error. Scale error shall not exceed the tolerances specified in Table 1.

3.5.8 Friction and position error. Friction and position error shall not exceed the tolerances specified in Table I.

3.5.9 Effect of lead length. When the lead length (NO. 20 AWG stranded copper wire) between the transmitter and indicator synchro stators is changed from 5 to 300 feet, the indication shall not change more than 1° of arc.

3.5.10 Leakage.

3.5.10.1 Hermetically sealed case. When A pressure of approximately zero psia is applied to the exterior of the transmitter case, the leak rate shall not exceed 0.10 micron cubic foot per hour when 10-percent helium is used in the filling medium and shall not exceed 1.0 micron cubic foot per hour when 100 percent helium is used as the filling medium.

3.5.10.2 Nonhermetically sealed case. There shall be no leakage when the transmitter is submerged in a partially filled container of water which has been evacuated to an air pressure of 1.0 psi.

3.6 Part numbering of interchangeable parts. All parts having the same manufacture's part number shall functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacture's part number and changes thereto.

3.7 Case. The case shall conform to figure 1 and shall provide either a hermetically sealed or gasket sealed enclosure for the internal mechanism of the transmitter. Sealing of the case shall be so accomplished that the seal will not be dependent upon material that would be affected by the action of any atmosphere to which the transmitter may be subjected.

3.7.1 Filling medium. When a hermetically sealed case is used, the filling

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medium shall be at least 98 percent purity free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall either a mixture of 88 to 92 percent nitrogen and the remainder helium or 100 percent helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.7.2 Pressure release. A blowout device shall be provided for releasing pressure in the case. The transmitter shall be so designed and constructed that if the pressure sensor connected to the pressure port leaks, the pressure in the case will be relieved and the escaping fluid will be discharged through the vent port.

3.8 Electronic components. Electronic components and the application thereof shall be in accordance with MIL-E-5400. Nonstandard parts shall not be used except with prior approval of the procuring activity.

3.9 Electrical connector. An MS33678C-14S-2P hermetically sealed electrical connector shall be provided.

3.10 Electrical characteristics. The transmitter synchro shall meet requirements of 3.10.1 through 3.10.3 with normal excitation of 26V 400-Hz single-phase power applied to the primary (rotor) fro a source having electrical characteristics conforming to MIL-STD-704 for a category B equipment.

3.10.1 Open secondary. The current drawn by the primary of the self-synchronous motor shall not exceed 240 ma with secondaries (stator) open circuited.

3.10.2 Power consumption. Power consumption of the transmitter synchro shall not exceed 6.24 Va.

3.10.3 Secondary voltage. The maximum open-circuit voltage measured between any two secondary leads shall be 11.8 ±0.3V. The variations between the three maximum secondary voltages on any one synchro shall not exceed 0.2V.

3.11 Wiring

3.11.1 External. The electrical receptacle shall be so wired that the 26V 400-Hz power supply will be connected to terminals A and B. External wiring between the transmitter and indicator shall be as shown on Figure 2.

3.11.2 Internal. Internal wiring of the transmitter shall be color coded and shall be connected to the receptacle pins as shown on figure 3. When gears as used to drive the transmitter counterclockwise for increasing values of the parameter being measured, terminals S1 and S3 shall be reversed internally to cancel the effects of the gear reversal to assure clockwise rotation of the indicator synchro shaft and pointer with increasing values.

Table I. Scale error tolerances.

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Tolerances (±) PSI

Scale Point PSI	Deg*	Room & High Temperature Friction Vibration	During & After Reliability High & Low Temperature Exposure Overpressure	High & Low Temperature	Pointer Oscillation	Position Error
Ø	2Ø	3.Ø	4.5	3.75	2.25	
5Ø	52	3.5	5.25	4.5	2.75	
1ØØ	84	6.Ø	5.Ø	5.Ø	3.Ø	
15Ø	116	4.5	5.75	5.5	3.5	
2ØØ	148	5.Ø	7.5	6.25	3.75	
25Ø	18Ø	5.5	8.25	7.Ø	4.25	5.Ø
3ØØ	212	6.Ø	9.Ø	7.5	4.5	
4ØØ	275	7.Ø	1Ø.5	.75	5.25	
5ØØ	34Ø	8.Ø	12.Ø	1Ø.Ø	6.Ø	

* Assumes indicating standard electrical zero per MIL-S-2Ø7Ø8.

3.12 Jewel bearings. When used, synthetic sapphire or ruby jewel bearings shall be in accordance with MIL-B-27497.

3.13 Finishes and protective coatings

3.13.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except the small holes, straight threads, case inserts, and parts in hermetically sealed enclosure need not be anorized.

3.13.1.1 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

3.13.1.2 When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.14 Weight. The weight of the transmitter shall not exceed 1.7 pounds.

3.15 Screw threads. Unless otherwise specified herein, the threads of all machine screws shall conform to MIL-S-7742.

3.16 Identification of product. Equipment, assemblies,, and parts shall be marked for identification in accordance with MIL-STD-13Ø.

3.16.1 Nameplate. A nameplate shall be permanently attached to the rear of the case or the required information shake be permanently etched on the side of the case where paint is not needed as protective coating. Where applicable, the information marked thereon shall conform to MIL-STD-13Ø. The nameplate shall not be glued or cemented to the case not shall metal decals be used. Etching shall be permanently inscribed so that it will withstand all normal environments for the life of the transmitter.

3.17 Workmanship. Workmanship shall be in accordance with MIL-STD-454,

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requirement 9.

3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operations, or performance of the transmitter, they shall be held or limited accordingly.

3.18 Cadmium. Cadmium is not to be used in the transmitter unless approved by the preparing activity. There shall be no cadmium used on the outside case of the transmitter. The fluid whose pressure is being measured by the transmitter shall not come in contact with any cadmium in the transmitter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as other wise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection and testing of transmitters shall be classified as follows:

a. Qualification Inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection condition. Unless otherwise specified, all inspections shall be performed in accordance with test conditions specified in the applicable test method document or applicable paragraph(s) in this specification.

4.3.1 Standard atmospheric conditions. Whenever this pressure a temperature existing at the time of the test are not specified definitely, the test shall be conducted at an atmospheric pressure of approximately 29.92 inches Hg and at an ambient temperature of 165° +5°, Ø°C. For tests conducted at room temperature the temperature shall be at 25° ±5°C.

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4.3.2 Test voltage. The transmitter shall be tested with 26V 400-Hz single phase power. Unless otherwise specified, the voltage and frequency shall be maintained with ±1 percent of the nominal value.

4.3.3 Tapping. The transmitter shall be lightly tapped or vibrated before each scale error test reading.

4.3.4 Test standard. The transmitter shall be tested against an indicating standard having maximum error of ±0.1 of a circular degree. The pressure source , such as a precision deadweight tester, shall have a maximum inherent error or ±0.03 psi.

4.4 Qualification inspection

4.4.1 Test samples. The test samples shall consist of a minimum of 10 transmitters representative of the production equipment. The sample shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Qualification inspection. The qualification test shall consist of all the test specified under 4.6 test methods

4.5 Quality conformance inspection. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual test. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product
b. Scale error
c. Friction error
d. Position error
e. Dielectric strength
f. Leakage
g. Electrical zero and verification of leads
h. Sensors matching (20 PSI Venting Pressure) See 4.6.24.1

4.5.2 Sampling tests. One transmitter shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual test
b. Low temperature
c. High Temperature
d. Vibration e. Electrical characteristics f. Damping tests (all paragraph 4.6.21) g. Sensors matching (see 4.6.24).

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4.5.2.1 Rejection and retest. When one transmitter selected from a production run fails to meet the specification, no transmitters on hand or later produced shall be accepted until the extent and cause of failure have been determined.

4.4.2.2 Individual tests may continue. For operational reasons, individual test may be continued pending the investigation of a sampling test failure. But final acceptance of transmitters on hand or later produced shall not be made until it is determined that items meet all requirements of this specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may existing transmitters already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test Methods

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, dimensions, and weight.

4.6.2 Scale error. The transmitter shall be tested for scale error at both normal operating and room temperatures at the test points specified in Table I for both upscale and down scale readings. The test points on the upscale reading shall be measured by bringing the input reading up to, but not exceeding, the specified test point. The difference in the scale error after vibration from the input value shall be brought down to, but not below, the specified test point.

4.6.3 Friction error. This test shall be conducted at both normal operating and room temperatures and may be performed concurrently with the scale error test. The difference in reading before vibrating the transmitter and after vibrating shall be the friction error which shall not exceed the tolerance specified in Table I.

4.6.4 Position error. This test may be conducted in conjunction with the test specified in 4.6.2 except at room temperature only. The transmitter shall be energized to obtain a reading of approximately one-half scale. The change in indication produced rotating the transmitter from normal operating position to position 90° clockwise and counterclockwise (longitudinal axis vertical) shall be noted. The maximum change in indication shall not exceed the tolerance specified for position error in table I.

4.6.5 Dielectric strength A potential 200V rms at commercial frequency shall be applied between any pin of the electrical connector, not connected to the case, and the case for a minimum period of 5 seconds. There shall be no insulation break-down. This test shall be conducted at room temperature.

4.6.6 Leakage

4.6.6.1 Hermetically sealed case. The hermetically sealed transmitter shall

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be tested for leaks at room temperature by means of a mass-spectrometer-type helium leak detector. The detected 1 rate for the 10-percent helium and 90-percent nitrogen filling medium shall not exceed 0.1 micron cubic foot per hour at pressure differential of 12 to 14.7 psi.

4.6.6.2 Nonhermetically sealed case. The Nonhermetically sealed transmitter shall be immersed in water in a container which shall be filled to not more than 70 percent of its capacity. The water level shall be 2 to 4 inches above the nearest part of the transmitter housing. The container shall first be evacuated to 0.50 psi air pressure and held at this pressure for 1 minute or until air bubbles cease to be given off by the liquid, whichever is greater. The Pressure shall then be raised to 1.0 psi and held at this pressure for a minimum period of 5 minutes. Any bubbles resulting from air or fluid leakage of entrapped air on the various exterior parts of the case shall not be considered as leaks.

4.6.7 Electrical zero and verification of terminal leads. This test shall be conducted at room temperature. With power applied to the transmitter per Figure 3, pressure shall be applied at the pressure connection of the transmitter with the vent connection open to ambient pressure. Electrical zero and verification of terminal leads shall be as follows:

a. The voltage from PIN C to PIN D (S3S1) shall begin at null and shall being phase with rotor voltage pin A to PIN B as the pressure is decreased fro approximately 250 psig to 0 psig.

b. The voltage between PIN B and PIN C shall be near a minimum and shall decrease to minimum as the pressure is increased from 250 psig to 296 psig.

c. Increasing pressure 250 psig to 296 psig shall be result in an increasing PIN C to PIN A voltage and decreasing PIN A to PIN D voltage.

4.6.7.1 Electrical zero adjustment. With pressure applied, the transmitter shall be subjected to , and shall meet, the test specified in 4.6.7 during which the zero adjustment shall be operated to the extreme positions. The required minimum adjustment shall be ±7 angular degrees from the electrical zero position.

4.6.8 Low temperature. The transmitter shall be subjected to a low temperature test in accordance with MIL-STD-810, method 502, at 54 ±2°C for a period of 4 hours after which it shall be subjected to the scale error and friction error test while still at that temperature. The errors shall not exceed the tolerances specified in Table I for low temperature.

4.6.9 High temperature. The transmitter shall be placed in a chamber and subjected to a temperature of 165° +5° -0° for a 4-hour period. Throughout this test, the transmitter shall be electrically connected to an appropriate indicator with power applied. At the end of the 4-hour period and with the transmitter still at specified temperature, the scale error and friction error shall not exceed the tolerances specified in table I.

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4.6.10 Vibration. The transmitter shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514, Equipment Category (b.1), figure 514.2-2, Curve L. At the conclusion of this test, the transmitter shall meet the test specified in 4.6.6.1 or 4.6.6.2, as applicable.

4.6.11 Electrical characteristics. The transmitter shall be tested to determine compliance with the requirements specified herein with respect to primary current, power input, secondary voltage, and operation of either one or two indicators with a change in indication of not more than 1° of arc.

4.6.12 Low temperature exposure. The transmitter shall be subjected to a low temperature exposure test in accordance with MIL-STD-810, method 502. The transmitter shall be exposed to a temperature of -62° ±2°C for a minimum period of 48 hours after which it shall be operated throughout its entire range. It shall exhibit no erratic characteristics during operation at the 62° ±2°C temperature. The temperature shall then be raised to -54° ±2°C and the transmitter maintained at this temperature for minimum of 24 hours. At the end of 24-hour period, the transmitter shall be subjected to the scale error and friction error test. The scale errors shall not exceed the tolerances specified in table I for high temperature exposure.

4.6.14 Altitude-temperature. The transmitter shall be subjected to an absolute pressure equivalent to an altitude of 85,00 feet for a period of 1 hour at s temperature of -54° ±2°C. While still at this pressure and temperature, the transmitter shall be subjected to the scale error test. The scale error shall not exceed the tolerances specified in table I for low temperature.

4.6.15 Electromagnetic interference. The transmitter shall be subjected to the electromagnetic interference test specified in MIL-STD-461 up to 400 Mhz for class ID equipment. The test methods shall be specified in MIL-STD-462.

4.6.16 Salt fog. The salt fog test shall be conducted in accordance with MIL-STD-810, method 509, at room temperature for a period of 48 hours. The transmitter shall be examined and there shall be no evidence of external corrosion or deterioration that would affect subsequent operation. The transmitter shall then meet the test specified in 4.6.5. Before the dielectric strength test is conducted, the external of the transmitter may be washed with distilled water and air dried.

4.6.17 Voltage and frequency. The transmitter shall be subjected to the scale error and friction tests during which the voltage and frequency shall be varied from 22 to 30V and 380 to 420Hz, or any combinations thereof. The scale and friction errors shall not exceed those specified in table I. This test shall be conducted at room temperature.

4.6.18 Effect of lead length. The lead length between the transmitter and indicator synchro stators shall be changed form 5 to 300 feet of No. 20 AWG stranded copper wire. The indication shall not change more than 1° of arc. This test shall be conducted at room temperature.

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4.6.19 Pressure bosses and connector strength. The pressure bosses and connector mountings shall be sufficiently rigid to withstand a force of 25 foot-pounds of bending moment applied in any direction perpendicular to the longitudinal axis of transmitter without the calibration shifting out of tolerance, distortion of the base plates, or evidence of damage to the pressure bosses, connector welds, or soldering. This test shall be conducted at room temperature and with the transmitter secured by the mounting flange. At the conclusion of this test, the transmitter shall be subjected to and meet the sealing test.

4.6.20 Overpressure (750 psi). This test shall be conducted at room temperature. A pressure of 750 psi shall be applied to the transmitter for 10 minutes. Not less than 1 hour following this test, the transmitter shall be subjected to the scale error test on increasing pressure only. The scale error shall not exceed the tolerances specified in table I for overpressure. Application of the 750-psi overpressure shall position the pointer of the indicating standard off the scale approximately midway between the 500- and 0-psi graduations as required for the overpressure stop in 3.4.1.8.

4.6.20.1 Overpressure (500 psi). This test shall be punctured or the seal shall be broken and a pressure of 1000 psi shall be applied at the input pressure port. Leakage of the internal mechanism through the hole in the case shall be cause for rejection. The sample transmitter need not be checked for accuracy following this test. The transmitter shall be then dissembled and inspected for compliance with 3.4.1.7.1, 3.4.1.7.2, 3.4.1.9, and 1.4.1.9.1.

4.6.20.2 Case pressure release. The pressure blowout device shall be tested either under simulated increasing case pressure or by actual injection of pressure into the transmitter case. The device shall operate at a pressure of 90 ± 30 psi relative to the atmosphere.

4.6.21 Damping. A 500 psi pressure shall be applied to the transmitter. A quick release valve shall be used to release the pressure such that the time interval for the pressure to decay from 500 psi to 100 psi shall be no more than 0.1 seconds. The corresponding time for the transmitter indication to change from 500 psi to 100 psi shall be 3 to 7 seconds. This test shall be conducted with oil, such as 1005 oil or equivalent, supplied to the pressure port of the transmitter and the oil pressure and trans mitter output monitored with a strip chart recorder. Neither restrictors nor a capillary line shall be used to connect the pressure input to the pressure element. This test shall be conducted tree times at -54 ±2°C. After the damping test the transmitter shall be tested and meet the requirements of 4.6.2 and 4.6.3.

4.6.22 Helium. The case shall be punctured or the filling tub cut on the case by means of a mass-spectrometer-type helium leak detector. Failure to detect helium leakage shall be cause for rejection.

4.6.23 Reliability

4.6.23.1 Test program. Prior to starting the reliability test the contractor

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shall prepare a detailed reliability test program in accordance with MIL-STD-781 and submit it to the procuring activity for formal approval. (See 6.3)

4.6.23.2 Test procedures. Each sample transmitter shall be subjected to a minimum of 1000 hours of operation consisting of a oil-line pressure oscillation at a frequency of 60 ±3 cpm fro 150 to 300 ±10 psi. Temperature cycling shall be in accordance with MIL-STD-781, Test Level J. Once in approximately each 16 hours the pressure cycling shall be stopped for a period of not longer than 1 minute during which the pressure shall be reduced to 0 psi before resuming the cycling. The scale error at room temperature test shall be conducted at least once every 200 hours. Throughout the pressure cycling, for 10 minutes of each hour the transmitter shall be subjected to sinusoidal vibration cycling in accordance with MIL-STD-810, Method 514, Figure 514.2-2, Curve H at 10g. After completion of this test, accuracy of each sample transmitter shall be within the tolerances specified in Table I for during and after reliability testing. During each accuracy check, a failure shall be counted for each sample transmitter that does not meet the specified tolerances. Accept-reject criteria shall be in accordance with MIL-STD-781, test plan I. No preventive maintenance shall be performed. Reliability test records shall also be in accordance with MIL-STD-781. (See 6.3)

4.6.24 Sensors matching. The transmitter adjustment shall be adjusted, and locked, for operation with the test indicator and the adjustment shall not be further disturbed for the balance of the sensor matching test. For each of the following conditions, the transmitter shall be tested and meet the scale error(sensor matching), at room temperature, at the test points specified in table I for both up-scale and down-scale readings. The test point on the up-scale readings shall be measured by bringing the input reading up to, but not exceeding, the specified test point. The down-scale error shall be measured in the same manner, except that the input reading value shall be brought down to, but not below, the specified test point.

4.6.24.1 20 PSI vent pressure. A 20 psi pressure shall be applied to the vent port connection of the transmitter. Pressures shall be applied to the pressure port connection which shall exceed that applied to the vent port connection by each of the test point listed in table I.

4.6.24.2 40 PSI vent pressure. A 40 psi pressure shall be applied to the vent port connection of the transmitter. Pressures shall be applied to the pressure port connection which shall exceed that applied to the vent port connection by each of the test points listed in table I.

4.7 Retention of qualification. Each vendor listed on the Qualified Products List (QPL) will be required to certify their product on DD Form 1718 every two years.

4.8 Inspection and test. Test methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

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5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C", as specified. (See 6.2)

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in quantity unit pack (QUP) of one each. Each item will be provided a preservation method IC-1 IAW MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Items shall be preserved IAW MIL-P-116 and MIL-STD-2073/1A to provide a method IC-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford protection against corrosion, deterioration, and physical damage during shipment from the source of supply to the first activity as specified in MIL-P-116 and MIL-STD-2073/1A.

5.2 Packing.

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-131, barrier material, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient cushioning/wrap material between the item and unit container to protect the unit container from punctures. Intermediate container requirements shall conform to PPP-B-636, water resistant, unless otherwise specified by the contracting activity. Intermediate containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak value in G's established for the item.

5.2.2 Level A. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-636, weather-resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with protection required. Closure shall be in accordance with appropriate PPP-B-636 procedures, as specified by contractor, special packaging instructions (SPI), and contracting activity.

5.2.3 Level B. Same as Level "A" unless otherwise specified by special packaging instructions (SPI) or the contracting activity.

5.2.4 Level C. Item will be packed in containers in such a manner as to afford adequate protection against physical damage during direct domestic

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shipment form the supply source to the first receiving activity. The packs shall conform to MIL-STD-2073/1A.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, special packaging instructions (SPI) and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging test shall be conducted IAW rough handling as in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

6.1 Intended use. The TRU-81/A transmitter covered by this specification is intended for use in transmitting, by electrical signal, the oil pressure in the lubricating system of aircraft engines to an indicator or indicators.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)
c. Levels of packaging and packing required and reinspection date marking (see section 5).

6.3 Data requirements. When this specification is used in an acquisition and data is required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contractor or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraphs No.	Data requirement title	Applicable DID No.	Option
3.4.1.9	Engineering drawings	DI-E-7031	------
4.6.23.1	Reliability test plan	DI-R-7033	------
4.6.23.2	Reliability test report	DI-R-7034	------

(Data item description related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.191, Vol II AMSDL. Copies of data item descriptions acquired by the contractors in connection with specific

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acquisition functions should be obtained from Defense Printing Service Detachment Office or as directed by the contracting officer.

6.4 Qualification considerations. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products Lists whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufactures are urged to arrange to have the products that they propose to offer the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is OC-ALC/TICLA, Tinker AFB, OK 73145-5990 and information pertaining to qualification of products ma be obtained from that activity.

6.5 Subject term (key word) listing.

Differential
Sender, pressure
Sensor, pressure
Synchro
Transducer, pressure
20 G

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE-99

Preparing Activity
AIR FORCE-71

Project Number:
6620-F497

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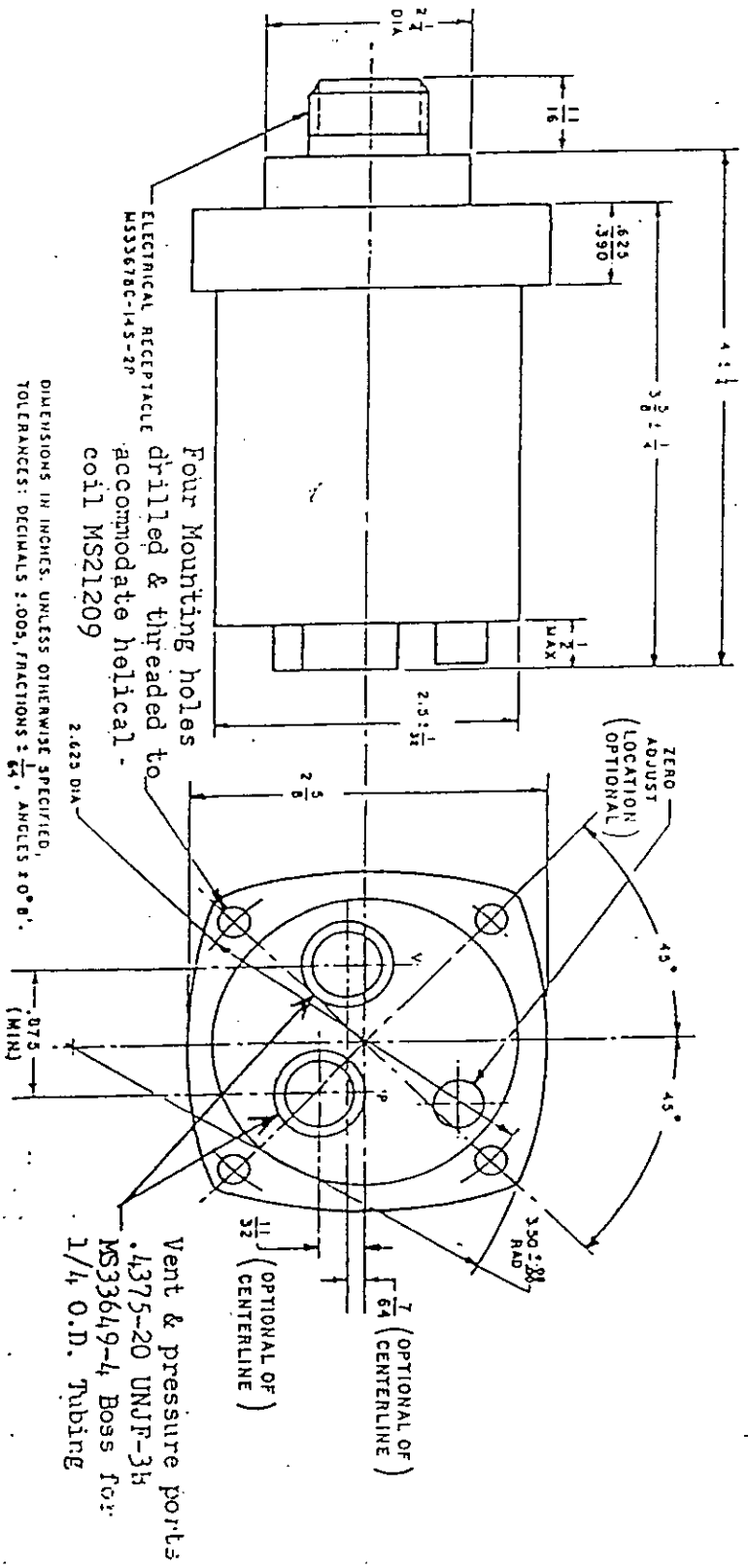


FIGURE 1. Case Dimensions.

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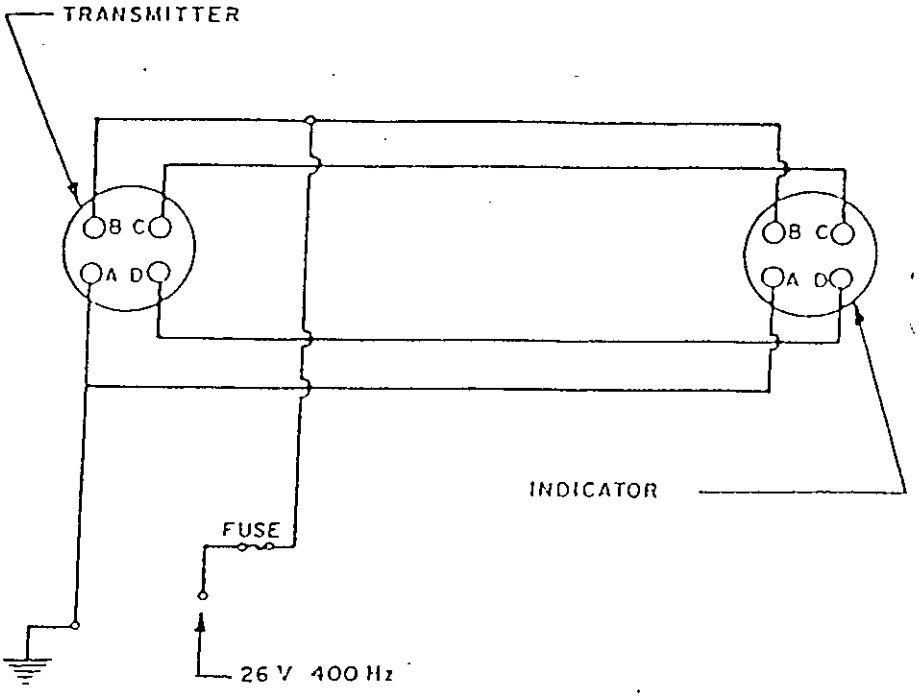


FIGURE 2. Wiring diagram - External, Single Indicator.

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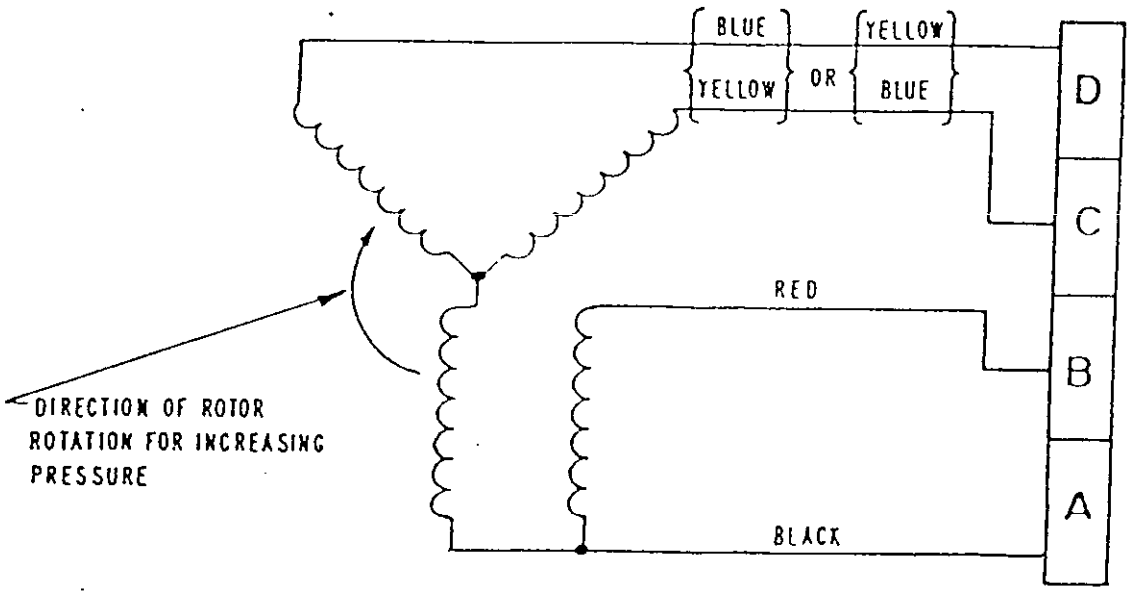


FIGURE 3. Internal Electrical Circuit.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-38394C(USAF)	30 MAY 1993

3. DOCUMENT TITLE

Transmitter, Pressure, TRU-81/A, Oil 0-500 PSI

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
QC-ALC/TICLA		
c. ADDRESS (Include Zip Code) Tinker AFB OK 73145-3037	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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